

2016.0 RANGE ROVER (LG), 415-01

# INFORMATION AND ENTERTAINMENT SYSTEM

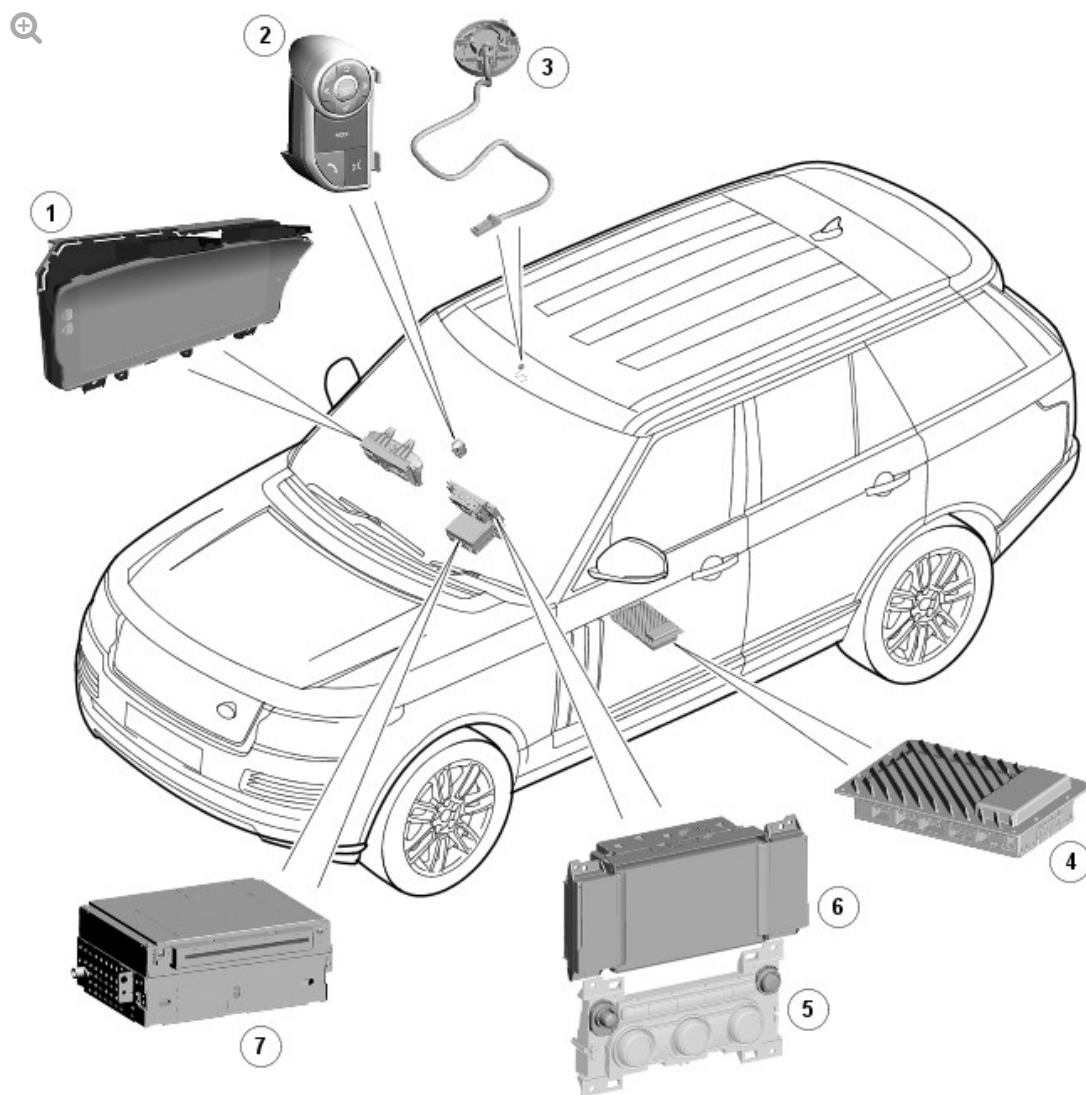
DESCRIPTION AND OPERATION

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COMPONENT LOCATION

BLUETOOTH® PHONE SYSTEM



E141837

| ITEM | DESCRIPTION                    |
|------|--------------------------------|
| 1    | IC (instrument cluster)        |
| 2    | Left Steering Wheel Switchpack |
| 3    | Microphone                     |
| 4    | AAM (audio amplifier module)   |
| 5    | ICP (integrated control panel) |
| 6    | TS (touch screen)              |
| 7    | IAM (integrated audio module)  |

## OVERVIEW

The Bluetooth® mobile phone system uses the customers own Bluetooth® capable mobile phone in conjunction with the vehicle information and entertainment system. The system is controlled by the IAM (integrated audio module). The IAM has a Bluetooth® antenna incorporated into the unit.

Mobile phone handsets must be paired with the IAM, requiring the input of a 4 digit PIN (personal identification number) before they can be used with the vehicle system.

Once paired, any mobile phone can be 'docked' to the vehicle without re-entering the PIN. In addition, the last connected device will dock automatically next time it is present in the vehicle and the ignition is in power mode 6 (ignition on) or power mode 7 (engine running).

Bluetooth® wireless technology allows electronic devices to communicate wirelessly through a short-range RF (radio frequency) network. Bluetooth® wireless technology can simultaneously handle data and voice transmissions which allows the vehicle's Bluetooth® phone system to provide hands free operation of the users Bluetooth® phone.

The Bluetooth® mobile phone system uses the MOST (media oriented system transport) ring for communication with the other components in the entertainment and information system.

The system comprises:

- IAM (integrated audio module)
- IC (instrument cluster) (for phone information in message center)
- TS (touch Screen)
- Microphone
- Left Steering Wheel Switchpack
- AAM (audio amplifier module)

Pairing of new mobile phones is achieved by selection of 'Search New' from

the device list screen. The system has two pairing modes; 'vehicle to device' and 'device to vehicle'. Up to 10 Bluetooth® devices can be paired to the vehicle.

**NOTE:**

There is no physical connection or cradle between the mobile phone and the IAM. Communication between the mobile phone and the vehicle system is only via Bluetooth®. This can limit the available functions dependant on the mobile phone used.

The system allows the driver to make, receive and end phone calls using the TS (touch screen) or the left steering wheel switchpack or voice control (where specified).

The Land Rover Advanced Voice system can be specified.

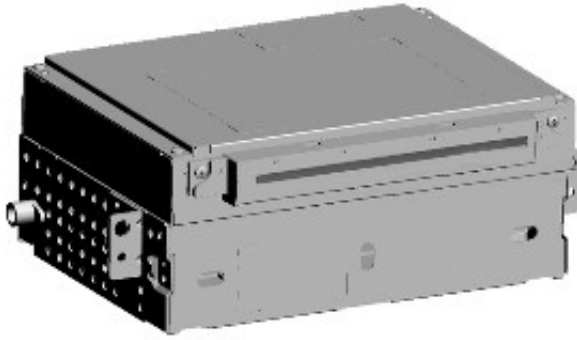
For additional information, refer to: [Voice Control](#) (415-01 Information and Entertainment System, Description and Operation).

Compatible Bluetooth® mobile phones can communicate with the vehicle's in-built phone system. Pairing and connection must be made with the ignition on or the engine running.

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## DESCRIPTION

### INTEGRATED AUDIO MODULE



E141841

The IAM (integrated audio module) is located in the instrument panel, behind the ICP (integrated control panel) and the ATC (automatic temperature control) module. The IAM communicates with the Bluetooth® mobile phone using internal hardware and a Bluetooth® antenna and software for phone functionality.

The IAM is connected to the MOST (media oriented system transport) ring and communicates with the TS (touch screen) and AAM (audio amplifier module) in order to provide the phone feature in the vehicle.

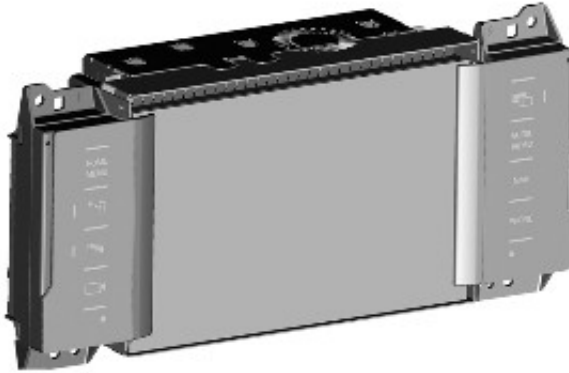
When the Bluetooth® mobile phone is specified, the IAM supports the Bluetooth™ mobile phone functionality and audio streaming. Bluetooth® mobile phone software and a Bluetooth® antenna are located within the IAM. The software within the IAM processes the Bluetooth® wireless signals from the users Bluetooth® mobile phone and allows audio output, voice input and data display on the TS.

The hands free microphone is hardwired to the IAM. The microphone is also used by the Land Rover Advanced Voice system for phone and other infotainment system operation.

Refer to the Audio section for further detail of the IAM.

For additional information, refer to: [Audio System](#) (415-01 Information and Entertainment System, Description and Operation).

## TOUCH SCREEN



E141842

The TS (touch Screen) provides the user interface of the Bluetooth® phone system.

The TS communicates with the rest of the audio/infotainment system on the MOST ring and allows control of the phone system and other infotainment systems from a single point.

The TS communicates with the IAM on the MOST ring and provides the driver interface and driver display of the phone system. The TS also provides driver display and control of the audio, the rear view camera, the navigation system, voice system, rear seat entertainment and other vehicle functions.

For additional information, refer to: [Audio System](#) (415-01 Information and Entertainment System, Description and Operation).

The phone system and other systems are operated by a combination of the soft key switches located in the TS switchpacks on each side of the TS and the 'virtual' soft keys displayed on the TS, in addition to the left steering wheel switchpack.

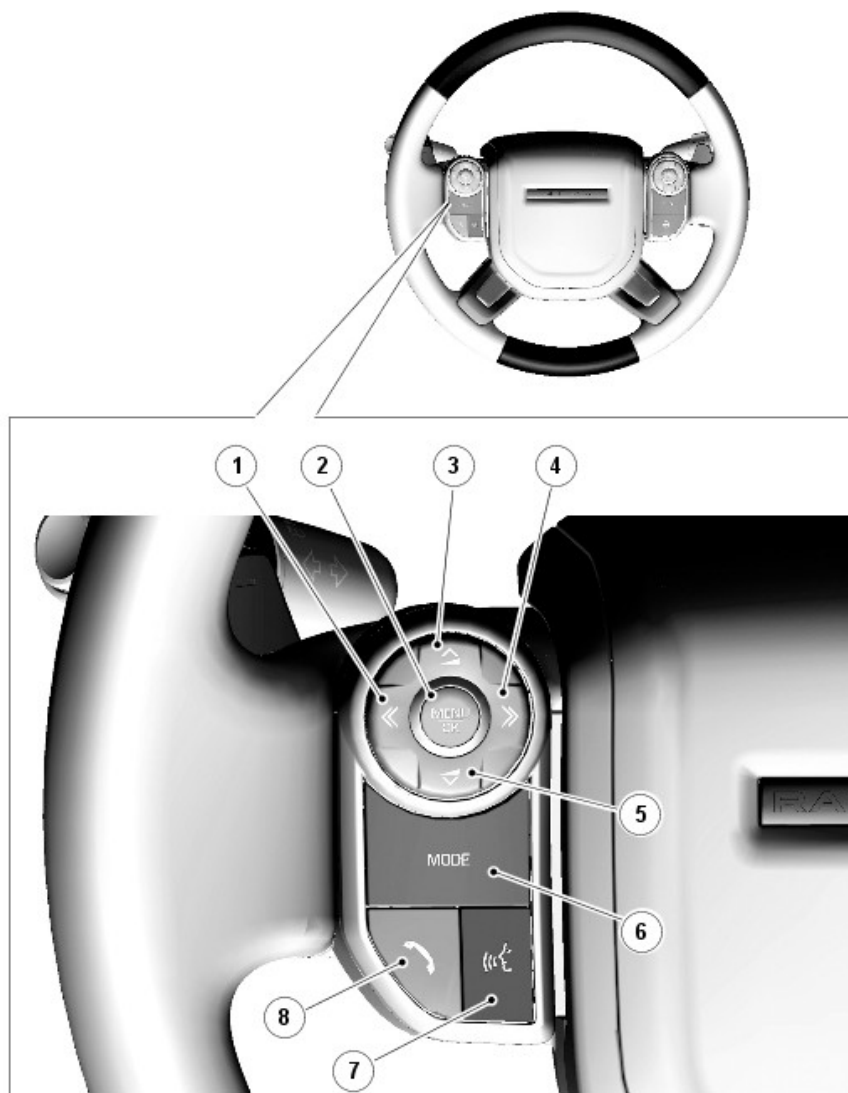
## MICROPHONE



E141843

A directional type microphone is fitted to the driver side of the overhead console so that it is directed towards the driver. It is connected to the IAM for hands free telephony, voice control and DVC (dynamic volume control) systems. The IAM has an integrated noise suppression and echo cancellation system for hands-free phone use.

#### LEFT STEERING WHEEL SWITCHPACK



#### ITEM

#### DESCRIPTION

|   |                                                                                                                                                                                                   |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Seek, previous radio preset, previous track, scroll down phone list, when 'Last 10' calls shown in TS will scroll forward through lists 'Last 10 Dialed', 'Last 10 Missed' and 'Last 10 Received' |
| 2 | Menu/OK button (instrument cluster menu control), answer an incoming telephone call or end an active telephone call                                                                               |
| 3 | Volume Increase, Scroll Up when in 'Phonebook' and 'Last 10' Screens                                                                                                                              |
| 4 | Seek, next radio preset, next track, scroll up phone list, when 'Last 10' shown in TS will scroll back through lists 'Last 10 Received', 'Last 10 Missed' and 'Last 10 Dialed'                    |
| 5 | Volume Decrease, Scroll Down when in 'Phonebook' and 'Last 10' Screens                                                                                                                            |
| 6 | Mode Button                                                                                                                                                                                       |
| 7 | Voice Button                                                                                                                                                                                      |
|   |                                                                                                                                                                                                   |



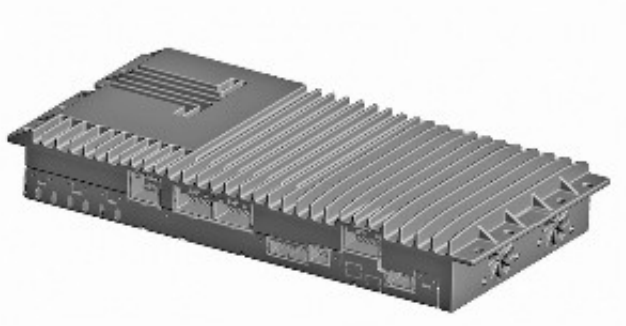
The switches for the phone system are located on the left steering wheel switchpack. The driver can use these switches for several phone related functions including:

- Accept incoming calls.
- Reject incoming calls.
- Make phone calls from the mobile telephones own phone book (phone book/call lists).

## AUDIO AMPLIFIER MODULE

### NOTE:

Meridian Signature Reference System audio amplifier module shown



E141845

The AAM (audio amplifier module) is located below the left front seat. It is connected to the audio system via the MOST. Speaker connections from the amplifier are hardwired.

## BLUETOOTH®

The Bluetooth® phone system allows the user to pair and connect their own mobile phone (Bluetooth® enabled only) to the vehicle phone system.

Once connected, the user can use the vehicle hands free functions. The Bluetooth® system limits the functions to those available in the Bluetooth® hands free profile.

The available features include:

- Make/receive calls.
- Dial voicemail number.
- Signal strength, network operator and battery level indication on the TS (if supported by the phone handset software).
- Phonebook download (if supported by the phone handset).
- Dialed calls (if supported by the phone handset).
- Missed calls (if supported by the phone handset).
- Received calls (if supported by the phone handset).

## **BLUETOOTH® PAIRING**

Before a Bluetooth® mobile phone can be used on the vehicle hands free system, the phone must be 'paired' to the Bluetooth® phone system. The following steps describe the 'pairing' process.

There are two ways of pairing a mobile phone to the vehicle system: From vehicle to device or from device to vehicle. Land Rover recommends customers to use device to vehicle pairing process.

### **NOTE:**

The process of pairing and connecting a mobile phone with the vehicle using the mobile phone will vary depending on the type of mobile phone used. Selecting 'search new' only appears if pairing for the 1st time, subsequent attempts to use the 'search new' soft key needs to be accessed by pressing the Change Phone 1st soft key.

- Switch the ignition system on (accessory power mode 4, ignition power mode 6 or engine running power mode 7) and ensure that the TS is active.
- From the home menu, select Phone.
- A menu will appear. Select Search new.

#### NOTES:

- The mobile phone must set to 'discoverable' for the pairing to work (vehicle to device only).
- The vehicle's Bluetooth® system is discoverable for only 3 minutes (device to vehicle only).

#### Device to Vehicle (Recommended Procedure)

- Select **Device** to vehicle option. On some mobile phones, this is referred to as new paired device. See the mobile phone operating instructions for further information.
- Using the mobile phone, search for the vehicle.
- When the vehicle's Bluetooth® system is discovered (named Range Rover), follow the on-screen instructions. Select **Yes** when prompted, to confirm the pairing. Either the mobile phone or the vehicle system will ask for a PIN (personal identification number). When prompted, enter a four digit PIN of your choice and select **OK** to confirm.
- Enter the same PIN into the other device.
- Once the mobile phone is paired and connected to the system, a confirmation message will be displayed before switching to the Digit Dial screen.

#### Vehicle to Device (Secondary Option)

- Select **Vehicle** to device option.

- Identify the mobile phone from the displayed list and select the corresponding Pair and Connect option.
- A PIN will be displayed in the TS, enter this PIN into the mobile phone.
- Once the mobile phone is paired and connected to the system, a confirmation message will be displayed before switching to the Digit Dial screen.

**NOTE:**

Some mobile phones require the Bluetooth® pairing to be set as 'authorized' or 'trusted' in order to automatically connect. Refer to the mobile phone's operating instructions for further information.

## CHANGING THE CONNECTED PHONE

Up to 10 Bluetooth® devices (mobile phones or other Bluetooth® audio devices) can be paired with the vehicle in the same way. However, only one can be connected and ready for use as a phone at any one time.

**NOTE:**

Bluetooth audio devices will use one of the 10 allocated pairing slots, but will not be displayed in the phone device list if they do not support telephony function.

To connect a different paired phone, follow the steps below:

- From the Home menu, select **Phone**.
- A menu will appear. Select **Change phone**.
- Identify and select the mobile phone from the displayed list.
- Once the mobile phone is connected to the system, a confirmation message will be displayed before switching to the Digit Dial screen.

## ADVANCED VOICE CONTROL SYSTEM

The advanced voice system provides the driver with the option of voice control for a range of supported functions. The following systems include Voice functionality:

- Radio, satellite radio and Digital Radio (if fitted).
- CD player.
- Hard disk drive/virtual CD player.
- USB and auxiliary connections.
- Navigation system.
- Bluetooth® phone system.
- Vehicle notepad.

For further details of the Advanced Voice Control system refer to the Voice Control section.

For additional information, refer to: [Voice Control](#) (415-01 Information and Entertainment System, Description and Operation).

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## OPERATION

### BLUETOOTH® PHONE SYSTEM

The Bluetooth® mobile phone system is controlled from the TS (touch screen), via voice control and the left steering wheel switchpack. Signals from the steering wheel switchpack are sent via the clockspring to the TS. The TS sends control signals on the MOST (media oriented system transport) ring to the IAM (integrated audio module), where these signals are sent via Bluetooth to the connected mobile phone. Speech is picked up by the microphone and sent to the IAM. Audio from the mobile phone is sent from the IAM on the MOST ring to the AAM (audio amplifier module) and is output on the vehicle speaker system.

The hands free profile of the system allows the display of the network signal

strength, network operator and phone battery level indication on the vehicle display, where the mobile phone supports this function. Also, if applicable, the mobile phone will show a "car" or "headset" symbol to indicate it is in hands free profile.

Mobile phone manufacturers continually update hardware and software to standard specifications laid down by the Bluetooth® Special Interest Group (SIG), which defines how Bluetooth® should work in an automotive environment. However, because different makes and models may use different software, not all mobile phones are fully compatible with Land Rover, but through testing individual phone handset models, Land Rover has produced a list of compatible phone handsets and the appropriate software version for Land Rover Bluetooth® connectivity.

Refer to the Bluetooth® Connectivity section of the Land Rover website at [www.landrover.com](http://www.landrover.com), for a list of compatible mobile phones. The list also describes how to check the software version of each individual phone manufacturer. This list must be consulted by users, sales and service teams to confirm whether customer's mobile phone is compatible or not. Performance will vary, based on the phone's software version, battery condition, coverage and the network provider. Mobile phones are warranted by their manufacturer, not Land Rover.

Land Rover is continually validating compatible mobile phone and software combinations. The most up-to-date compatibility list will always be available on-line from Land Rover.

**NOTE:**

To achieve full Bluetooth® mobile phone functionality, it is crucial that the mobile phone software level matches the version detailed in the list of compatible phone handsets.

**BLUETOOTH®**

Bluetooth® is a short-range RF (radio frequency) technology that operates

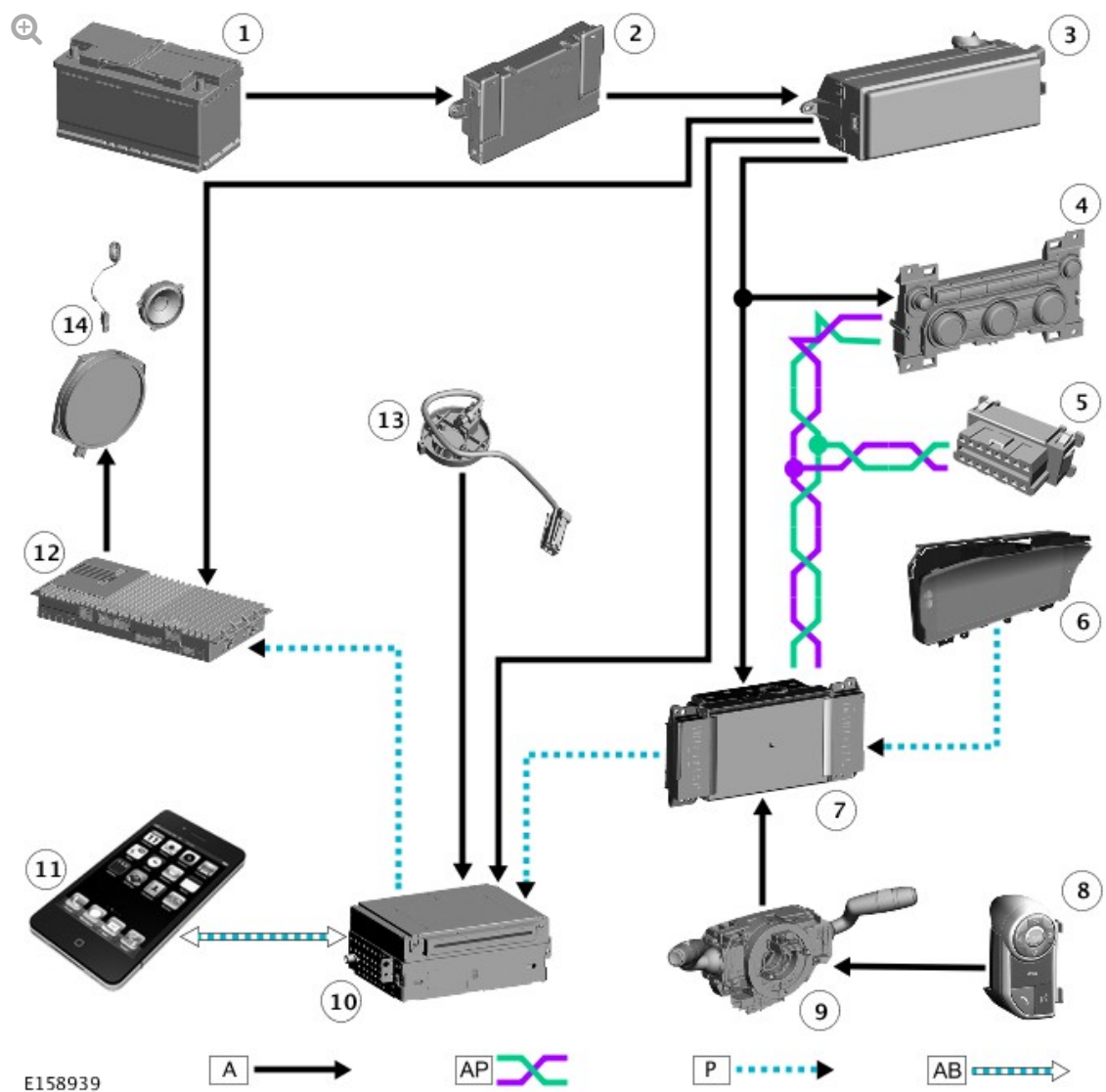
at 2.4 GHz and is capable of transmitting voice and data wirelessly. The effective range of Bluetooth® devices is 32 feet (10 meters) with a data transfer rate of 1 Mbps.

Bluetooth® is a wireless RF connection which operates with the user's own mobile phone. It does not have to be fixed into the vehicle and is designed to function without an external aerial. The mobile phone can be located anywhere within the cabin, even in a bag, or jacket pocket. It can work from the luggage compartment, although the signal could be compromised. The mobile phone can be charged from the 12 volt power socket or USB (universal serial bus) (if supported) while in use.

The Bluetooth® system supports Bluetooth® Hands-Free Profile (HFP), Phonebook Access Profile (PBAP), Advanced Audio Distribution Profile (A2DP) and Audio Video Remote Control Profile (AVRCP).

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## CONTROL DIAGRAM



## ITEM

## DESCRIPTION

|   |                                                                             |
|---|-----------------------------------------------------------------------------|
|   | A = Hardwired; AP = Medium Speed CAN Comfort Bus; P = MOST; AB = Bluetooth® |
| 1 | Battery                                                                     |
| 2 | BJB (battery junction box)                                                  |
| 3 | RJB (rear junction box)                                                     |
| 4 | ICP (integrated control panel)                                              |
| 5 | Diagnostic Socket                                                           |
| 6 | IC (instrument cluster)                                                     |
| 7 | TS (touch screen)                                                           |
| 8 | Left steering wheel switchpack                                              |



|    |                               |
|----|-------------------------------|
| 9  | Clockspring                   |
| 10 | IAM (integrated audio module) |
| 11 | Mobile phone                  |
| 12 | Microphone                    |
| 13 | AAM (audio amplifier module)  |
| 14 | Vehicle speakers              |

Refer to the Audio section for full MOST control diagram.

For additional information, refer to: [Audio System](#) (415-01 Information and Entertainment System, Description and Operation).